Do Now · retrieval

Lesson 081 · Merge Sort: Bottom-up Merge Tracing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a sorted run?

Q2

How does insertion sort work?

Q3

What is O(n log n) complexity?

Q4

Binary search sorted [1,3,5,7,9,11,13] for 11: show midpoint index sequence.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Merge sort divides the input into smaller runs; merging two already sorted runs repeatedly selects the smallest available front value.

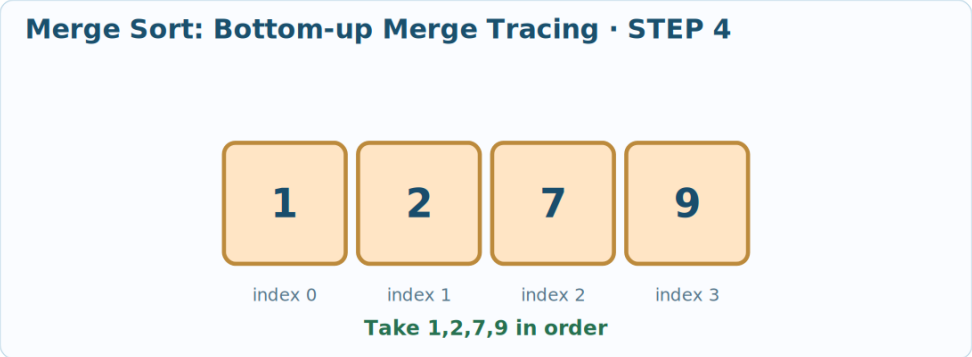
Core idea 2

Apply — This bottom-up variant starts with single-item runs and doubles run width; it needs no recursive calls. The top-down variant recursively splits to length 0/1, then merges on return; Lesson 186 connects the two.

Core idea 3

Evaluate — Worst-case time O(n log n); standard array merging uses O(n) extra space. Include comparisons, not just the final sorted sequence.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

merge sort — Sort by merging sorted runs.

run — Consecutive already sorted segment.

merge — Combine two sorted sequences.

auxiliary space — Extra memory beyond input.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

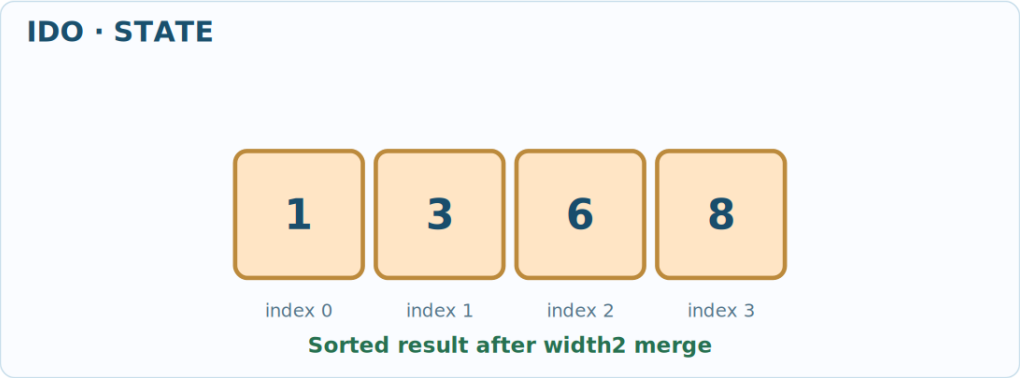
Problem

Bottom-up merge [8,3,6,1]; show pair results and full result.

Follow the worked method

1. Method and result — Pairs [3,8] and [1,6] → [3,8,1,6]; merge fronts → [1,3,6,8].

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Method and result
Pair sort [2,5,4,7]; merge sorted runs → [2,4,5,7].

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Given [7,2,9,1], state array after width1 pair merges and final width2 merge. [6]

Question 2

Merge sorted runs [2,7] and [1,9] and list chosen values in order, with comparisons. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare bottom-up merge with insertion sort for large random data. Give worst time and auxiliary-space cost. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Merge sort requires no additional array storage in its standard array implementation.

Use key vocabulary from memory

1. Explain merge sort in your own words.

2. Explain run in your own words.

3. Explain merge in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.